THEORY ASSIGNMENT – 3

JavaScript | Deadline: 22 October 2026

1. What is the difference between frontend and backend JavaScript?

Frontend JavaScript runs in the user's browser and controls webpage behavior, such as buttons, forms, and animations. Backend JavaScript runs on a server, commonly using Node.js, and handles tasks such as APIs, databases, and authentication.

2. What is DOM?

DOM stands for Document Object Model. It represents an HTML page as a tree of objects, allowing JavaScript to read and change webpage content, structure, and styles.

3. What is an event in JavaScript?

An event is an action or occurrence detected by the browser, such as a mouse click, key press, form submission, or page load. JavaScript can run a function in response to an event.

4. What is the event loop in JavaScript?

The event loop helps JavaScript handle asynchronous tasks. It checks whether the call stack is free and then moves ready callbacks from queues to the stack so they can run.

5. What is hoisting in JavaScript?

Hoisting is JavaScript's behavior of processing declarations before executing the code. Function declarations can be called before their definition; var declarations are hoisted and initialized as undefined, while let and const cannot be used before initialization.

6. What is event bubbling?

Event bubbling is when an event starts on the target element and then travels upward through its parent elements. For example, clicking a button inside a div can trigger click handlers on both the button and the div.

7. What is the difference between named, anonymous, and arrow functions?

A named function has a name, for example: function greet() {}. An anonymous function has no function name, for example: function() {} when used as an expression. An arrow function uses =>, for example: const greet = () => {}; arrow functions also handle 'this' differently and do not have their own this.

8. Explain for-of and for-in loops.

A for-of loop iterates over values in an iterable, such as an array or string: for (const value of [10, 20]) {}. A for-in loop iterates over enumerable property keys, commonly an object's keys: for (const key in person) {}.

9. Explain any five array methods.

1. push(): adds an item to the end of an array. 2. pop(): removes the last item. 3. shift(): removes the first item. 4. unshift(): adds an item to the beginning. 5. map(): creates a new array by applying a function to each item.

10. What is a callback function?

A callback is a function passed as an argument to another function so it can be called later. Example: setTimeout(() => { console.log('Hello'); }, 1000); the arrow function is the callback.